Adrian J. Duszkiewicz, PhD

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CURRENT AND PREVIOUS POSITIONS

Lecturer in Neuroscience *02.2026 – present*
University of Manchester, UK

Senior Postdoctoral Researcher *08.2022 – 01.2026*
University of Stirling, UK
Supervisor: Prof. Paul Dudchenko

Postdoctoral Researcher / Postdoctoral Fellow *11.2016 – 07.2022*
McGill University, Canada
Supervisor: Dr. Adrien Peyrache
Supported by EMBO Long-Term Postdoctoral Fellowship
and Sir Henry Wellcome Postdoctoral Fellowship

Postdoctoral Researcher *09.2015 – 10.2016*
University of Edinburgh, UK
Supervisor: Prof. Richard Morris

Graduate Research Assistant *09.2014 – 08.2015*
University of Edinburgh, UK
Supervisor: Prof. Richard Morris

EDUCATION

University of Edinburgh, UK *09.2011 – 06.2016*
PhD at the Centre for Cognitive and Neural Systems
Thesis: *Optogenetic dissection of the dopaminergic circuitry involved in memory consolidation*
Advisors: Prof. Richard Morris, Prof. Emma Wood

University of Edinburgh, UK *09.2007 – 07.2011*
BSc (Hons) Biological Sciences (Physiology), First Class
Thesis: *Thalamocortical circuit abnormalities in SAP-102 knockout mice*
Advisor: Prof. Peter Kind

RESEARCH GRANTS, FELLOWSHIPS AND AWARDS

Simons Initiative for the Developing Brain (University of Edinburgh): *04.2025 – 03.2026*
Collaborator. Primary investigator: Dr. Alfredo Gonzalez-Sulser.
Mechanisms mediating absence seizures across rat models of autism

Sir Henry Wellcome Postdoctoral Fellowship, Wellcome Trust: *05.2018 – 04.2022*
Functional dissection of the head-direction system

Long-Term Postdoctoral Fellowship, EMBO: *06.2017 – 04.2018*
Thalamocortical processing in the head-direction system

Edinburgh Neuroscience Travel Grant: learning to combine single unit recording with optogenetics in Prof. Naoshige Uchida's laboratory, Harvard University, USA *12.2012*

PhD Fellowship and *In Vivo* Science Strategic Skills Award *09.2011 – 08.2014*
Medical Research Council

The Physiological Society's Undergraduate Prize *06.2011*

Society for Endocrinology Summer Studentship *06.2010 – 08.2010*

TEACHING AND SUPERVISION

Co-supervisor of PhD student projects: Anuj Vadher (University of Stirling), Anna Rastedt (University of Edinburgh), Trisha Narayanan (University of Edinburgh), Noah Moore (University of Edinburgh) *2019 – 2026*

Co-supervisor of undergraduate research projects, MSc projects and internships: Eleanor Brown (co-author on paper), Jordan Shanel, Eliott Owczarek (co-author on paper), Pierre Orhan (co-author on paper). Laetitia Wang, Emily Gittings, Brenda Shen, Dapang Wang, Thalia McKenzie, Sean Van Doeselaer, Anna Rastedt, Ana Hart-Fernandez *2012 – 2026*

Guest lecturer - Neurobiology of Learning and Memory course for 4th year undergraduate students (University of Edinburgh) *2024 – 2025*

Tutor - Research Fundamentals course tutorials for 1st year postgraduate medical students (McGill University) *2018 – 2020*

Guest lecturer - Neurobiology of Cognition course for 4th year students (University of Edinburgh) *2014*

Tutor - General Neuroscience course tutorials for 4th year students (University of Edinburgh) *2012 – 2014*

Demonstrator - Brain and Behaviour course for 3rd year students (University of Edinburgh) *2012 – 2013*

SCIENTIFIC SERVICE

Peer Review: Nature Communications, Current Biology, eLife, Journal of Neuroscience, Journal of Neuroscience Methods *2020 – present*

Grant Review: Wellcome Trust, BBSRC, French National Research Agency *2022 - 2024*

Spring Hippocampal Research Conference symposium. Organizer, speaker and co-chair of session: *Hippocampal and Cortical Circuits in Rat Models of Neurodevelopmental Disorders* (incoming) *05.2025*

Spring Hippocampal Research Conference symposium. Organizer, speaker and co-chair of session and associated data blitz: *The Head-Direction Cortex: Circuits and Dynamics of Spatial Orientation* *05.2023*

ATTENDED COURSES

Flatiron CCN Neural Data Analysis Workshop (San Diego, CA) *11.2025*

NeuroMatch Academy NeuroAI Course (virtual) *07.2025*

NeuroMatch Academy Deep Learning Course (virtual) *07.2024*

EMBO Lab Leadership Course, Heidelberg, Germany *04.2022*

PEER-REVIEWED PUBLICATIONS

Duszkiewicz AJ[±], Burgalossi A, Fricker D, Peyrache A[±] (2025) The postsubiculum as a head-direction cortex. *Trends in Neurosciences*, 48: 829-840. [±]corresponding authorship

Mehrotra D*, Levenstein D*, **Duszkiewicz AJ,** Skromne Carrasco S, Booker SA, Kwiatkowska A, Peyrache A (2024) Hyperpolarization-activated currents drive neuronal activation sequences in sleep. *Current Biology*, 34: 3043–3054 *equal contribution

Duszkiewicz AJ[±], Orhan P, Skromne Carrasco S, Brown EH, Owczarek E, Vite GR, Wood ER, Peyrache A[±] (2024) Local origin of excitatory–inhibitory tuning equivalence in a cortical network. *Nature Neuroscience*, 27:782-792 [±]corresponding authorship

Duszkiewicz AJ*, Rossato JI*, Moreno A, Takeuchi T, Yamasaki M, Genzel L, Spooner P, Canals S, Morris RGM (2023) Execution of new trajectories toward a stable goal without a functional hippocampus. *Hippocampus*, 33(6): 769:786 *equal contribution

Peyrache A, **Duszkiewicz AJ**, Viejo G, Angeles-Duran S (2019) Thalamocortical processing of the head-direction sense. *Progress in Neurobiol*ogy 183: 101693

Duszkiewicz AJ, McNamara CG, Takeuchi T, Genzel L (2019) Novelty and dopamine-dependent memory consolidation: a tale of two systems. *Trends in Neurosciences*, 42(2):102-114 (>300 citations via Google Scholar)

Takeuchi T*, **Duszkiewicz AJ*,** Sonneborn A***,** Spooner P, Yamasaki M, Watanabe M, Smith CC, Fernandez G, Deisseroth K, Greene RW and Morris RGM (2016) Locus coeruleus and dopaminergic consolidation of everyday memory. *Nature*, 537:357-362 *equal contribution

Takeuchi T, **Duszkiewicz AJ** and Morris RGM. (2014) The synaptic plasticity and memory hypothesis: encoding, storage and persistence. *Philosophical Transactions of the Royal Society B: Biological Sciences* 369(1633):20130288

Wacker DW, Tobin VA, Noack J, Bishop VR, **Duszkiewicz AJ**, Engelmann M, Meddle SL and Ludwig M (2010) Expression of early growth response protein 1 in vasopressin neurones of the rat anterior olfactory nucleus following social odour exposure. *Journal of Physiology*. 558(23):4705-4717

PREPRINTS

Krausse S, **Duszkiewicz AJ**, Neftci E, Sommer ET, Renner A. Parallax error indicates simple cue-anchoring in the head-direction system. BioRxiv: 2025.04.25.650191

Moore N*, **Duszkiewicz AJ***, Asiminas A, Dudchenko PA, Peyrache A, Wood ER. Altered developmental profile of the cortical head-direction signal in a rat model of Fragile X Syndrome. BioRxiv: 2025.01.09.63213 *equal contribution

Orhan P, **Duszkiewicz AJ**, Peyrache A. Signature of random connectivity in the distribution of neuronal tuning curves. BioRxiv: 2022.08.30.505888

OTHER PUBLICATIONS

Peyrache A and **Duszkiewicz AJ** (2023) Memory and the rodent thalamus. Book chapter in Usrey WM and Sherman SM (Eds.): The Cerebral Cortex and Thalamus. Oxford University Press

Peyrache A and **Duszkiewicz AJ** (2021) A spatial map out of place. *Cell Research,* 31(6):605:606

Duszkiewicz AJ and Peyrache A (2017) Primates have sharp memories too. *Current Biol*ogy 27:R57-R76

OPEN-SOURCE DATASETS

Duszkiewicz AJ and Skromne Carrasco S (2024) Large-scale recordings of head-direction cells in mouse postsubiculum. DANDI Archive: 10.48324/dandi.000939/0.240327.2229.

SELECTED INVITED PRESENTATIONS

Origin of inhibitory tuning in the rodent head-direction cortex. Part of the German Neuroscience Meeting (NWG) symposium "*Neuronal representation of space, directions and goals in insects and vertebrates"*. Göttingen, Germany *03.2025*

Head-direction circuit - a new model system to study cortical function and dysfunction? University of Copenhagen, Denmark *11.2023*

The rings of power: using the head-direction system to probe cortical function in health and disease. Nencki Institute for Experimental Biology, Warsaw, Poland *10.2022*

Using the rodent head-direction system to reveal the principles of cortical inhibitory tuning. Young Investigator Seminar Series, FMI Basel, Switzerland (virtual) *08.2022*

The many faces of inhibition in the head-direction system. Centre for Integrative Physiology, University of Tübingen, Germany *04.2022*

Thalamic reticular nucleus exerts inhibitory control on the head-direction signal. MRC Brain Dynamics Unit, Oxford University, UK *07.2019*

Persistence of memory: new insights into the role of dopamine in memory consolidation. Nencki Institute for Experimental Biology, Warsaw, Poland *05.2019*

Memories built to last: new insights into dopaminergic control of memory consolidation. Bernstein Center for Computational Neuroscience, Berlin, Germany *09.2017*

Memories built to last: new insights into dopaminergic control of memory consolidation. Max Planck Institute for Biological Cybernetics, Tübingen, Germany *09.2017*

Locus coeruleus controls hippocampal memory consolidation through non-canonical release of dopamine. Part of the *"Role of the Locus Coeruleus in Cellular and Systems Consolidation"* symposium at the EBBS meeting, Bilbao, Spain *09.2017*

Optogenetic dissection of the dopaminergic circuitry involved in memory consolidation and behavioural tagging. Donders Discussions Graduate Student Conference, Donders Institute, Nijmegen, The Netherlands *11.2017*

Locus Coeruleus and dopamine-dependent memory consolidation. Radboud University, Nijmegen, The Netherlands *11.2017*

Neuromodulatory mechanisms behind behavioural tagging. Massachusetts Institute of Technology, Cambridge, USA *03.2013*

Mnemonic functions of dopamine in the hippocampus. Harvard University, Cambridge, USA *03.2013*

CONFERENCE ABSTRACTS

Duszkiewicz AJ, Rastedt AA, Vadher A, Wood ER, Peyrache A, Dudchenko PA (2024) Impaired multisensory integration within the head-direction circuit in a rat model of Fragile X Syndrome. Society for Neuroscience: abstract and poster

Rastedt AA, **Duszkiewicz AJ**, Kind PC, Dudchenko PA, Wyllie DJA, Wood ER (2024) Hippocampal place cell activity in a rat model of GRIN2B-related neurodevelopmental disorder. Society for Neuroscience: abstract and poster

Vadher A, Agnihotri M, Sebastian J, **Duszkiewicz AJ,** Wood ER, Peyrache A, Dudchenko PA (2024) Characterization of visual and self-motion cue-based navigation in a rat model of Fragile X Syndrome. Society for Neuroscience: abstract and poster

Skromne Carrasco S, Viejo G, Mannarelli R, Wang Y, **Duszkiewicz AJ**, Peyrache A (2024) Sharp waves of the head-direction system. Society for Neuroscience: abstract and poster

Duszkiewicz AJ, Skromne Carrasco S, Orhan P, Brown E, Owczarek E, Vite GR, Wood ER, Peyrache A (2022) Cortical inhibitory tuning reflects the frequency components of locally encoded features. FENS Forum: abstract and poster

Duszkiewicz AJ, Skromne Carrasco S, Orhan P, Brown E, Owczarek E, Vite GR, Wood ER, Peyrache A (2022) Cortical inhibitory tuning reflects the frequency components of locally encoded features. COSYNE meeting: abstract and poster

Duszkiewicz AJ, Wood ER, Peyrache A (2021) Non-uniform tuning in the cortical head-direction system. Society for Neuroscience – Global Connectome: abstract and poster

Duszkiewicz AJ, Wood ER, Peyrache A (2020) Locally non-uniform tuning in the cortical head-direction system. FENS Forum of Neuroscience: abstract and poster

Duszkiewicz AJ, Brown E, Wood ER, Peyrache A (2019) Inhibitory control in the head-direction system. Society for Neuroscience: abstract and poster

Duszkiewicz AJ, Brown E, Wood ER, Peyrache A (2018) Thalamic reticular nucleus fine-tunes the head-direction signal. Society for Neuroscience: abstract and poster

Duszkiewicz AJ, Brown E, Yamasaki M, Watanabe M, Peyrache A (2018) Thalamic reticular nucleus controls the gain of the head-direction signal. 2nd Interdisciplinary Navigation Symposium iNAV: abstract and poster

Duszkiewicz AJ, Brown E, Yamasaki M, Watanabe M, Peyrache A (2017) Thalamic processing of the head direction signal. Society for Neuroscience: abstract and poster

Duszkiewicz AJ*, Rossato JI*, Moreno A, Takeuchi T, Genzel L, Canals S, Morris RGM (2017) The role of the hippocampus in goal-directed navigation. Canadian Association of Neuroscience Meeting: abstract and poster. * equal contribution

Takeuchi T*, **Duszkiewicz AJ*,** Yamasaki M, Tse, D**,** Spooner P, Watanabe M, Deisseroth K, Fernandez G and Morris RGM. (2015) Catecholaminergic enhancement of initial memory consolidation in mice. Society for Neuroscience: abstract and poster. * equal contribution

Duszkiewicz AJ, Takeuchi T, Spooner P, Deisseroth K, Fernandez G and Morris RGM. (2015) Catecholaminergic neurons in mouse locus coeruleus are more strongly activated by novelty than catecholaminergic neurons in the ventral tegmental area. Society for Neuroscience: abstract and poster

Takeuchi T*, **Duszkiewicz AJ*,** Tse, D**,** Spooner P, Deisseroth K, and Morris RGM. (2015) Catecholaminergic control of initial memory consolidation in mice. BNA Festival of Neuroscience: abstract and poster. * equal contribution

Duszkiewicz AJ, Takeuchi T, Spooner P, Deisseroth K, and Morris RGM. (2015) Optogenetically identified catecholaminergic neurons in ventral tegmental area and locus coeruleus are activated

by novelty. BNA Festival of Neuroscience: abstract and poster

Duszkiewicz AJ, Takeuchi T, Deisseroth K and Morris RGM. (2014) Activity of optogenetically identified dopaminergic neurons in the ventral tegmental area during exploration of familiar and novel environments. FENS Forum of Neuroscience Meeting: abstract and poster

Takeuchi T, **Duszkiewicz AJ,** Tse, D**,** Spooner P, Deisseroth K, and Morris RGM. (2014) Selective catecholaminergic control of initial memory consolidation in mice. FENS Forum of Neuroscience Meeting: abstract and poster